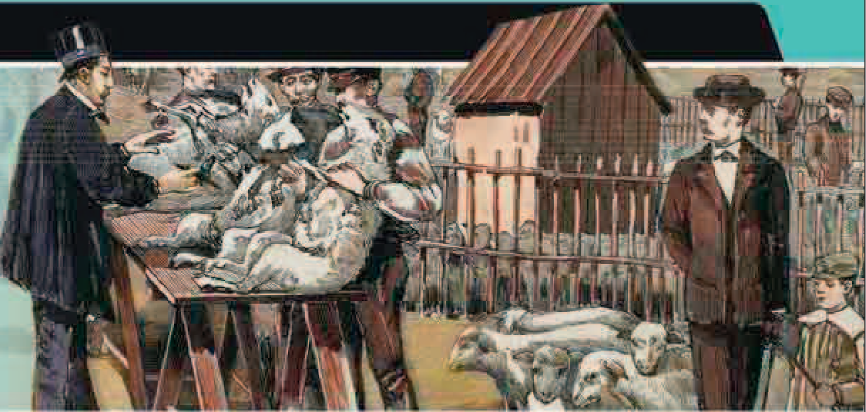
1880–81

“…THE **ANTHRAX VACCINE** THAT **FIRST SPREAD** THROUGH THE PUBLIC MIND **FAITH IN** THE **SCIENCE OF MICROBES.**”

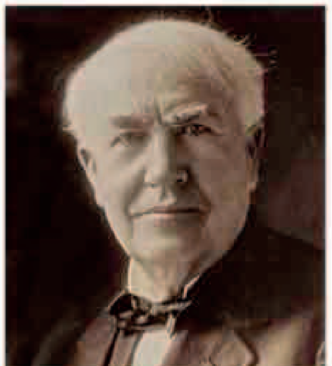
Emile Duclaux, French chemist and microbiologist, from *Pasteur: The History of a Mind*, 1896



This engraving depicts French bacteriologist Louis Pasteur vaccinating sheep against anthrax at Pouilly-le-Fort, France, in 1881.

in air to be 186,327 miles per second (299,864 kmps). This estimate matched the prediction of British physicist James Clerk Maxwell (see 1872–73).

German engineer **Werner von Siemens** (1816–92) demonstrated the **first electric locomotive** using an external power source at the 1879 Berlin trade fair.



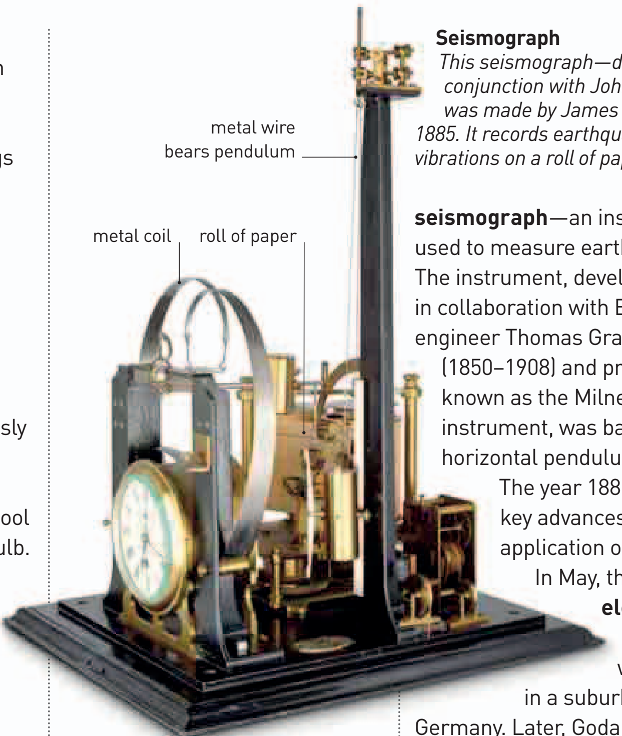
THOMAS ALVA EDISON
(1847–1931)

American inventor Thomas Edison is credited with many inventions, particularly in the field of telecommunications. He filed more than a thousand patents in the US, and others around the world. Edison applied the idea of mass production and teamwork to science and developed the world's first industrial research laboratory—his greatest invention.

IN 1880, BRITISH LOGICIAN AND PHILOSOPHER John Venn (1834–1923) developed the concept of the **Venn diagram**, which represents sets of things as circles, with overlapping circles indicating the subsets they have in common. Venn wrote about this concept in *Symbolic Logic*, which was published in 1881.

In February 1880, **Thomas Alva Edison** rediscovered a phenomenon that had previously been observed by others. He noticed that electricity would flow from a hot filament to a cool metal plate in an evacuated bulb. Edison patented this concept, which came to be known as the **Edison effect**. The electricity could flow only one way, so the setup acted like a valve to control the flow of current in the same way that a valve in a pipeline controls the flow of water. This concept became the basis of the valves used to amplify electrical signals in television and radio before the invention of the transistor.

The understanding of electricity was also aided by the **discovery of the piezoelectric effect** by French scientists **Pierre Curie** (1859–1906) and **Paul-Jacques Curie** (1856–1941). They discovered that voltage can be produced by applying pressure to a suitable material.



Seismograph
This seismograph—designed in conjunction with John Milne—was made by James White in 1885. It records earthquake vibrations on a roll of paper.

Conversely, applying an electric potential to crystals can make them vibrate at a very precise frequency. This effect has many applications, including driving the vibrating crystals in quartz clocks and watches.

Known as the father of modern seismology, British geologist **John Milne** (1850–1913), was teaching science and engineering in Japan when he became interested in earthquakes. In 1880 he was instrumental **in inventing the seismograph**—an instrument used to measure earthquakes. The instrument, developed in collaboration with British engineer Thomas Gray (1850–1908) and properly known as the Milne-Gray instrument, was based on a horizontal pendulum design.

The year 1881 saw two key advances in the application of electricity. In May, the **first electric tramway** was opened in a suburb in Berlin, Germany. Later, Godalming, UK, became the first town to have its **streets lit by electricity**.

In September 1881, German gynecologist **Ferdinand Adolf Kehrer** (1837–1914) carried out the **first modern cesarian section operation**. Both mother and baby survived.

French microbiologist **Louis Pasteur made a vaccine for anthrax** by using the oxidizing agent potassium dichromate to weaken its bacteria. He began using British surgeon Edward Jenner's term "vaccine" to refer to all such artificially weakened disease organisms. Jenner had coined the term in reference to smallpox (see 1796).

German scientist **Paul Ehrlich** (1854–1915)—whose cousin, pathologist Karl Weigert, (1845–1904) had been the first person to stain bacteria with dyes in the 1870s—found a more effective dye, **methylene blue**. This made it easier to identify and investigate bacteria, and was used by German physician Heinrich Koch to discover the bacteria that causes tuberculosis (see 1882–83).

First electric tramway
Developer of the first electric train, Werner von Siemens also worked on the first electric tram—the Gross-Lichterfelde in Germany.

1879 Thomas Alva Edison files for a patent for a **light bulb**

1879 German biologist Walther Flemming coins the terms **chromatin** and **mitosis**

1880 John Milne invents the **seismograph**

1880 Pierre and Paul-Jacques Curie discover **piezoelectricity**—electricity generated due to pressure

February 13, 1880 Thomas Edison rediscovers the **Edison effect**, also known as thermionic emission

1881 John Venn publishes *Symbolic Logic*, introducing the concept of **Venn diagrams**

May 16, 1881 The world's **first electric tramway** opens in Germany

1881 Alfred Russel Wallace proposes that the **Cambrian period** began about 28 million years ago

1881 Louis Pasteur develops a vaccine for **anthrax**

September 1881 Godalming, UK, becomes the first town to have its streets illuminated by **electric light**

September 25, 1881 Ferdinand Kehrer performs the first modern **cesarean section**

1881 Paul Ehrlich uses **methylene blue** to stain bacteria